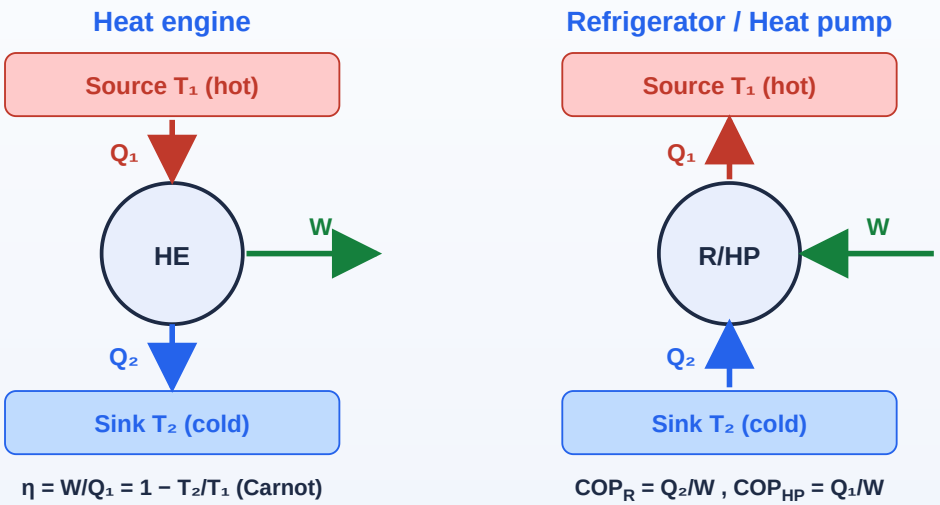
UPSSSC ALERT

UPSSSC JE

MECHANICAL THERMODYNAMICS

Complete English Edition

Unit 4 · 5 Marks · 2026



$\eta = W/Q_1 = 1 - T_2/T_1$ (Carnot)

$COP_R = Q_2/W$, $COP_{HP} = Q_1/W$

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UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **Thermodynamics**. Syllabus: properties of pure substances, steam tables, steam formation, wet and superheated steam, dryness fraction, degree of superheat, first law, steady flow process, SFEE, second law, heat source, sink and reservoir, heat engine, heat pump and refrigerator.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (P. K. Nag, R. K. Rajput, R. S. Khurmi).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Pure Substances & Properties of Steam

Pure Substances & Properties of Steam – Basic Concept

A **pure substance** has a uniform chemical composition in all phases — e.g. a mixture of water, ice and steam. Steam is the working fluid of power plants, so steam formation, its states (wet, dry saturated, superheated) and the use of **steam tables** must be understood.

1. Key Points

- **Steam formation (at constant pressure):** (i) heating water to saturation temperature — sensible heat h_f; (ii) evaporation — latent heat h_{fg} (temperature constant); (iii) superheating — $c_{ps}(T_{sup} - T_s)$.
- **Saturation temperature:** the boiling temperature at a given pressure; it rises with pressure (100°C at 1.013 bar).
- **Wet steam:** steam carrying water droplets; dryness fraction x = mass of dry steam/total mass. $h = h_f + x\, h_{fg}$, $v = x\, v_g$.
- **Dry saturated steam:** $x = 1$; $h = h_g = h_f + h_{fg}$. **Superheated steam:** steam heated above saturation temperature; $h_{sup} = h_g + c_{ps}(T_{sup} - T_s)$.
- **Degree of superheat:** $T_{sup} - T_s$. Superheated steam behaves nearly like an ideal gas; $c_{ps} \approx 2.1$ kJ/kg·K.
- **Critical point:** 220.9 bar and 374.15°C; here $h_{fg} = 0$ — water flashes directly into steam. **Triple point:** 0.006113 bar, 0.01°C — all three phases coexist.
- **Steam tables:** saturated steam tables (pressure- or temperature-based: T_s, v_g, h_f, h_{fg}, h_g, s_f, s_{fg}, s_g) and superheated steam tables. Reference: u and s of saturated water at 0.01°C are zero.
- **Calorimeters for dryness fraction:** bucket, throttling, separating, and combined separating-throttling calorimeter (for very wet steam).
- **Mollier (h–s) chart:** used to read enthalpy drops directly, especially in isentropic expansion of steam.
- **Other pure substances:** water–steam, air (in gas phase, homogeneous mixture), ammonia. An oil–water mixture is not a pure substance.

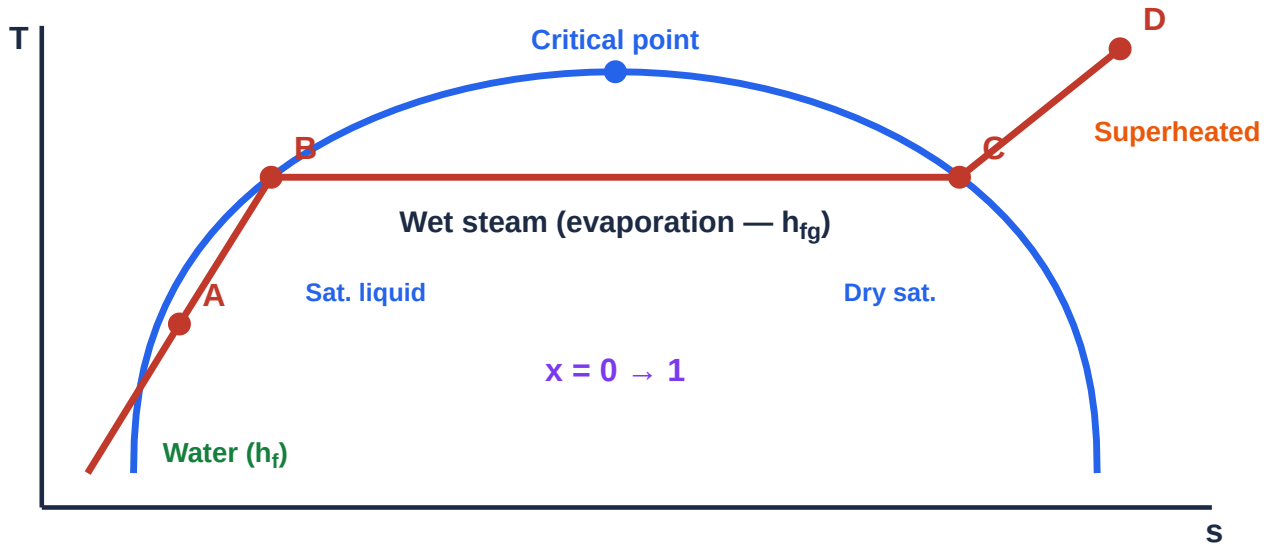


Fig 4.1: Steam formation on T–s diagram (A–B water, B–C evaporation, C–D superheating)

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Enthalpy of wet steam	$h = h_f + x\, h_{fg}$
Dry saturated steam	$h_g = h_f + h_{fg}$
Superheated steam	$h = h_g + c_{ps}(T_{sup} - T_s)$
Specific volume of wet steam	$v = x\, v_g$
Entropy of wet steam	$s = s_f + x\, s_{fg}$
Internal energy	$u = h - pv$
Dryness fraction	$x = m_g / (m_g + m_f)$
Throttling calorimeter	$h_{f1} + x\, h_{fg1} = h_{g2} + c_{ps}(T_{sup} - T_{s2})$
Combined calorimeter	$x = x_1 \times x_2$
Degree of superheat	$T_{sup} - T_s$

3. Detailed Explanation

- **T–s and p–v diagrams:** the saturated liquid line (x = 0) and saturated vapour line (x = 1) meet at the critical point to form a dome; inside is wet steam, left is compressed liquid, right is superheated steam.
- **Effect of increasing pressure:** T_s and h_f increase, h_{fg} decreases to zero at the critical point; h_g rises up to about 30 bar and then falls.
- **Principle of throttling calorimeter:** throttling wet steam to low pressure (h constant) makes it superheated; measuring temperature gives h_2, and $h_1 = h_2$ gives x. Steam must not be too wet ($x > \approx 0.93$).

- **Advantages of superheated steam:** higher heat content, less condensation and erosion in turbines, higher efficiency; hence modern plants use superheated steam.
- **External work of evaporation:** $p(v_g - v_f) \approx p\, v_g$; internal latent heat = $h_{fg} - p\, v_g$.

4. Comparison Table — Wet, Dry Saturated and Superheated Steam

Basis	Wet	Dry saturated	Superheated
Dryness fraction	$x < 1$	$x = 1$	Not applicable
Temperature	T_s	T_s	$T > T_s$
Enthalpy	$h_f + x\, h_{fg}$	h_g	$h_g + c_{ps}\Delta T$
Behaviour	With water droplets	Invisible pure vapour	Like an ideal gas

5. Solved Numerical Problems

Example 1. Steam at 10 bar has dryness 0.9 (h_f = 762.6, h_{fg} = 2015.3 kJ/kg). Find the enthalpy.
Solution:

1. $h = 762.6 + 0.9 \times 2015.3 = 762.6 + 1813.8 =$ **2576.4 kJ/kg**

Example 2. Steam at 10 bar (T_s = 179.9°C, h_g = 2777.1) is superheated to 250°C; c_{ps} = 2.1. Find degree of superheat and enthalpy.
Solution:

1. Degree of superheat = 250 − 179.9 = **70.1°C**
2. $h = 2777.1 + 2.1 \times 70.1 =$ **2924.3 kJ/kg**

Example 3. 2 kg of wet steam contains 0.2 kg of water. Find the dryness fraction.
Solution:

1. $x = 1.8/2 =$ **0.9**

Example 4. Steam at 5 bar (v_g = 0.3749 m³/kg) has x = 0.8. Find its specific volume.
Solution:

1. $v = x\, v_g = 0.8 \times 0.3749 =$ **0.300 m³/kg**

Example 5. In a separating-throttling calorimeter x_1 = 0.95 and x_2 = 0.97. Find the actual dryness fraction.
Solution:

1. $x = x_1 \times x_2 = 0.95 \times 0.97 =$ **0.9215**

6. Frequently Asked Facts

- Critical point: 220.9 bar, 374.15°C; latent heat is zero there.
- Wet steam $h = h_f + x\, h_{fg}$; superheated $h = h_g + c_{ps}(T_{sup} - T_s)$.
- Combined separating-throttling calorimeter for very wet steam.

- Latent heat decreases as pressure increases.

7. Note (Confusing Facts)

- For dry saturated steam x = 1, but x is not defined for superheated steam (it does not exceed 1).
- Temperature stays constant during evaporation even though heat is supplied (latent heat).
- The triple point temperature is 0.01°C, not 0°C.

Remember: One line: x = 0 saturated water, x = 1 dry saturated steam. Assertion–Reason: (A) Superheated steam is used in turbines. (R) It reduces blade erosion and increases efficiency. — Both true; R explains A.

30 Most Important One-Liner Facts — Pure Substances & Properties of Steam

1. Pure substance?

— Uniform chemical composition in all phases.

2. Saturation temperature at 1.013 bar?

— 100°C.

3. Symbol of sensible heat?

— h_f.

4. Symbol of latent heat?

— h_{fg}.

5. Temperature during evaporation?

— Constant.

6. Dryness fraction?

— Mass of dry steam ÷ total mass.

7. x for saturated water?

— 0.

8. x for dry saturated steam?

— 1.

9. Enthalpy of wet steam?

— $h_f + x\ h_{fg}$.

10. Enthalpy of superheated steam?

— $h_g + c_{ps}(T_{sup} - T_s)$.

11. Degree of superheat?

— $T_{sup} - T_s$.

12. c_{ps} of superheated steam?

— ≈ 2.1 kJ/kg·K.

13. Critical pressure?

— 220.9 bar.

14. Critical temperature?

— 374.15°C.

15. h_{fg} at critical point?

— Zero.

16. Triple point temperature of water?

— 0.01°C.

17. Specific volume of wet steam?

— $x\ v_g$.

18. Entropy of wet steam?

— $s_f + x\ s_{fg}$.

19. h_{fg} with rising pressure?

— Decreases.

20. T_s with rising pressure?

— Increases.

21. Calorimeter for very wet steam?

— Combined separating-throttling.

22. Principle of throttling calorimeter?

— $h_1 = h_2$.

23. x in combined calorimeter?

— $x_1 \times x_2$.

24. Name of h–s chart?

— Mollier diagram.

25. Reference state in steam tables?

— Saturated water at 0.01°C.

26. State inside the dome?

— Wet steam.

27. Right of the dome?

— Superheated steam.

28. External work of evaporation?

— p v_g (approx.).

29. Superheated steam behaves like?

— Ideal gas.

30. Internal energy (steam)?

— u = h − pv.

Practice MCQs — Pure Substances & Properties of Steam

Q1. Enthalpy of wet steam:

(a) $h_f + h_{fg}$

(b) $h_f + x\ h_{fg}$

(c) $x\ h_f + h_{fg}$

(d) $h_g + x\ h_{fg}$

✔ **Answer: (b) $h_f + x\ h_{fg}$**

Explanation: Fraction x is evaporated.

Q2. Latent heat at the critical point:

(a) Maximum

(b) Zero

(c) 2257 kJ/kg

(d) Infinite

✔ **Answer: (b) Zero**

Explanation: Liquid–vapour distinction vanishes.

Q3. Dryness fraction of dry saturated steam:

(a) 0

(b) 0.5

(c) 1

(d) More than 1

✔ **Answer: (c) 1**

Explanation: x = 1.

Q4. Critical pressure of water:

(a) 100 bar

(b) 220.9 bar

(c) 1.013 bar

(d) 374 bar

✔ **Answer: (b) 220.9 bar**

Explanation: 220.9 bar.

Q5. Latent heat of evaporation as pressure increases:

(a) Increases

(b) Decreases

(c) Remains constant

(d) First decreases then increases

✔ **Answer: (b) Decreases**

Explanation: Zero at the critical point.

Q6. To measure dryness of very wet steam we use:

(a) Throttling calorimeter
(b) Combined separating-throttling calorimeter
(c) Bomb calorimeter
(d) Junker's calorimeter

✔ **Answer: (b) Combined separating-throttling calorimeter**

Explanation: Throttling alone cannot superheat it.

Q7. Triple point temperature of water:

(a) 0°C
(b) 0.01°C
(c) 4°C
(d) 100°C

✔ **Answer: (b) 0.01°C**

Explanation: 0.01°C, 0.006113 bar.

Q8. Enthalpy of superheated steam:

(a) $h_g + c_{ps}(T_{sup} - T_s)$
(b) $h_f + c_{ps}T$
(c) $h_{fg} + c_{ps}\Delta T$
(d) $x\ h_g$

✔ **Answer: (a) $h_g + c_{ps}(T_{sup} - T_s)$**

Explanation: Superheat added.

Q9. 2 kg wet steam with 0.4 kg water — dryness fraction:

(a) 0.2
(b) 0.6
(c) 0.8
(d) 0.4

✔ **Answer: (c) 0.8**

Explanation: 1.6/2.

Q10. Temperature during evaporation:

(a) Increases
(b) Decreases
(c) Remains constant
(d) Uncertain

✔ **Answer: (c) Remains constant**

Explanation: Latent heat.

Q11. The h–s chart is called:

(a) Mollier diagram
(b) T–s diagram
(c) Psychrometric chart
(d) p–v diagram

✔ **Answer: (a) Mollier diagram**

Explanation: Mollier.

Q12. Specific volume of wet steam is approximately:

(a) v_f
(b) $x\ v_g$
(c) v_g/x
(d) $v_g + v_f$

✔ **Answer: (b) $x\ v_g$**

Explanation: v_f is negligible.

Q13. Throttling calorimeter works on:

(a) Constant entropy
(b) Constant enthalpy
(c) Constant temperature
(d) Constant volume

✔ **Answer: (b) Constant enthalpy**

Explanation: $h_1 = h_2$.

Q14. At 10 bar T_s = 180°C, steam at 230°C — degree of superheat:

(a) 50°C
(b) 230°C
(c) 180°C
(d) 410°C

✔ **Answer: (a) 50°C**

Explanation: 230 − 180.

Q15. Which is not a pure substance?

(a) Water and steam
(b) Ice and water
(c) Oil and water
(d) Ammonia

✔ **Answer: (c) Oil and water**

Explanation: Heterogeneous, different composition.

SSC JE / UPSSSC JE Previous Year Questions — Pure Substances & Properties of Steam

Q1. Dryness fraction of steam is defined as:

SSC JE (Previous Year)

(a) Mass of dry steam/total mass
(b) Mass of water/total mass
(c) Dry steam/water
(d) Water/dry steam

✔ **Answer: (a) Mass of dry steam/total mass**

Explanation: Definition.

Q2. Temperature of water at the critical point:

SSC JE (Previous Year)

(a) 100°C
(b) 273°C
(c) 374.15°C
(d) 500°C

✔ **Answer: (c) 374.15°C**

Explanation: 374.15°C.

Q3. Temperature of superheated steam is:

SSC JE (Previous Year)

(a) Below saturation temperature
(b) Equal to saturation temperature
(c) Above saturation temperature
(d) 0°C

✔ **Answer: (c) Above saturation temperature**

Explanation: Definition.

Q4. In throttling of steam, ____ remains constant: SSC JE (Previous Year)

(a) Enthalpy
(b) Entropy
(c) Pressure
(d) Temperature

✔ **Answer: (a) Enthalpy**

Explanation: $h_1 = h_2$.

Q5. The point where all three phases coexist: SSC JE (Previous Year)

(a) Critical point
(b) Triple point
(c) Boiling point
(d) Freezing point

✔ **Answer: (b) Triple point**

Explanation: Triple point.

Q6. Enthalpy of dry saturated steam: SSC JE (Previous Year)

(a) h_f
(b) h_{fg}
(c) $h_f + h_{fg}$
(d) $h_f - h_{fg}$

✔ **Answer: (c) $h_f + h_{fg}$**

Explanation: h_g.

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Chapter 2: Basic Concepts & First Law of Thermodynamics

Basic Concepts & First Law of Thermodynamics – Basic Concept

Thermodynamics is the science of energy, heat and work, their inter-conversion and their relation to the properties of matter. The **first law** is the law of conservation of energy — energy can neither be created nor destroyed, only converted. Engines, turbines, compressors and nozzles are analysed with it.

1. Key Points

- **Types of system:** closed (energy but no mass transfer), open (both mass and energy) and isolated (neither). An open system is called a control volume.
- **Property:** intensive (independent of mass: pressure, temperature, density) and extensive (depends on mass: volume, energy, enthalpy). Extensive ÷ mass = specific property (intensive).
- **Thermodynamic equilibrium:** thermal (equal temperature), mechanical (equal pressure) and chemical equilibrium together.
- **Zeroth law:** if two systems are each in thermal equilibrium with a third, they are in equilibrium with each other — the basis of thermometry.
- **Heat and work:** both are boundary phenomena and path functions, not properties; their differentials are inexact. Sign convention: heat supplied to the system +, work done by the system +.
- **First law (closed system):** $Q = \Delta U + W$; for a cycle $\oint \delta Q = \oint \delta W$. Internal energy (U) is a property.
- **Enthalpy:** $H = U + pV$; for an ideal gas $h = c_p T$ and $u = c_v T$. $c_p - c_v = R$ and $\gamma = c_p/c_v$ (1.4 for air).
- **Perpetual motion machine of the first kind (PMM-I):** a machine producing work continuously without energy input is impossible — first law.
- **Steady flow process:** properties inside the control volume do not change with time; mass flow in = mass flow out.
- **SFEE:** $h_1 + C_1^2/2 + gz_1 + q = h_2 + C_2^2/2 + gz_2 + w$ (per kg); flow work pv is included in enthalpy.

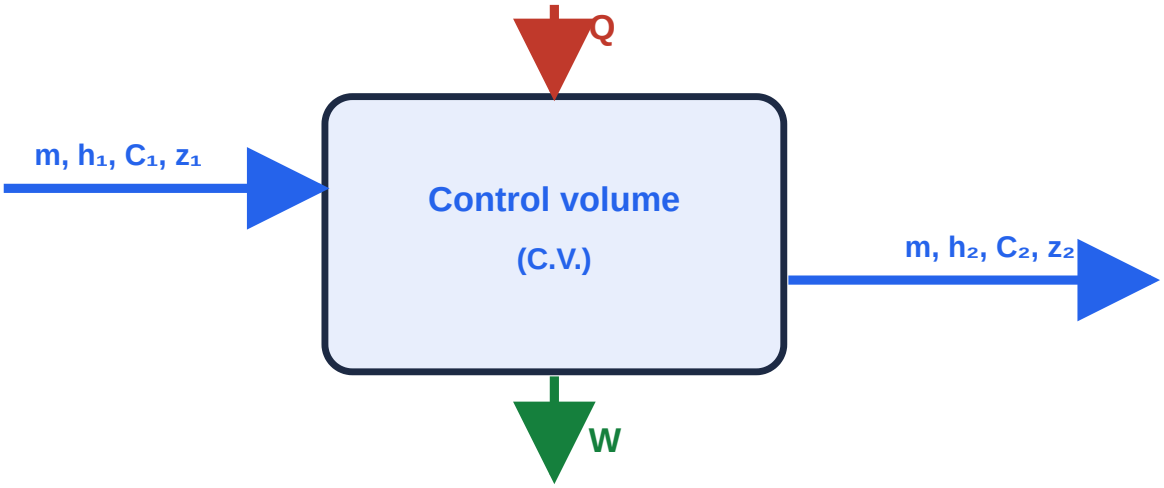


Fig 4.2: Steady Flow Energy Equation (SFEE)

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
First law (closed)	$Q = \Delta U + W$
Cycle	$\oint \delta Q = \oint \delta W$
Enthalpy	$H = U + pV$
Ideal gas	$pV = mRT$, $c_p - c_v = R$
Constant volume work	$W = 0$, $Q = mc_v\Delta T$
Constant pressure work	$W = p(V_2 - V_1)$, $Q = mc_p\Delta T$
Isothermal work	$W = p_1V_1 \ln(V_2/V_1)$
Adiabatic ($pV^\gamma = C$)	$W = (p_1V_1 - p_2V_2)/(\gamma - 1)$
Polytropic ($pV^n = C$)	$W = (p_1V_1 - p_2V_2)/(n - 1)$
SFEE	$h_1 + C_1^2/2 + gz_1 + q = h_2 + C_2^2/2 + gz_2 + w$
Nozzle exit velocity	$C_2 = \sqrt{[2(h_1 - h_2)]}$ (C_1 negligible)

3. Detailed Explanation

- **Summary of processes:** constant volume (isochoric) – zero work; constant pressure (isobaric); isothermal – for ideal gas $\Delta U = 0$, $Q = W$; adiabatic – $Q = 0$; throttling – $h_1 = h_2$ (isenthalpic).
- **Applications of SFEE:** turbine ($q \approx 0$, ΔKE, ΔPE negligible): $w = h_1 - h_2$; compressor: $w = -(h_2 - h_1)$; nozzle ($q = w = 0$): $C_2 = \sqrt{(2\Delta h)}$; boiler/condenser ($w = 0$): $q = h_2 - h_1$; throttle valve: $h_1 = h_2$.
- **Quasi-static process:** an infinitely slow process with the system always near equilibrium; only such a process can be drawn as a line on a p–V diagram and its work is $\int p\, dV$.

- **Free expansion:** expansion into vacuum — W = 0, Q = 0, so ΔU = 0; the temperature of an ideal gas stays the same, but the process is irreversible.
- **Joule's experiment:** paddle-wheel work equalled the heat produced, establishing the equivalence of heat and work (1 cal = 4.186 J).

4. Comparison Table — Closed vs Open System

Basis	Closed system	Open system
Mass transfer	No	Yes
Energy transfer	Yes	Yes
First law	Q = ΔU + W	SFEE
Examples	Closed piston–cylinder	Turbine, compressor, nozzle

4. Comparison Table — Point vs Path Function

Basis	Point function (property)	Path function
Depends on	End states only	Path of process
Differential	Exact	Inexact
Examples	p, V, T, U, H, S	Heat, work

5. Solved Numerical Problems

Example 1. A closed system receives 100 kJ of heat and does 40 kJ of work. Find the change in internal energy.

Solution:

1. ΔU = Q − W = 100 − 40 = **60 kJ** (increase)

Example 2. 1 kg of air is compressed at a constant 1 bar from 0.8 m³ to 0.5 m³. Find the work.

Solution:

1. $W = p(V_2 - V_1) = 100 \times (0.5 - 0.8) =$ **−30 kJ**
2. Negative sign: work done on the system

Example 3. Steam enters a turbine at 3200 kJ/kg and leaves at 2600 kJ/kg; flow 2 kg/s. Find the power (neglect heat loss, KE, PE).

Solution:

1. $w = h_1 - h_2 = 600$ kJ/kg
2. P = 2 × 600 = **1200 kW**

Example 4. Enthalpy drop in a nozzle is 180 kJ/kg; inlet velocity negligible. Find the exit velocity.

Solution:

1. $C_2 = \sqrt{(2 \times 180 \times 1000)} = \sqrt{360000} =$ **600 m/s**

Example 5. 0.1 m^3 of air at 10 bar expands isothermally to 1 bar. Find the work.

Solution:

1. $W = p_1V_1 \ln(p_1/p_2) = 1000 \times 0.1 \times \ln 10$
2. $W = 100 \times 2.303 =$ **230.3 kJ**

6. Frequently Asked Facts

- Nozzle from SFEE: $C_2 = 44.72\sqrt{\Delta h}$ (Δh in kJ/kg).
- Enthalpy remains constant in throttling.
- Heat and work are path functions; internal energy and enthalpy are properties.
- Work is zero in a constant volume process.

7. Note (Confusing Facts)

- Work is zero in free expansion even though volume increases.
- In an isothermal process heat is supplied yet temperature does not change (Q = W for ideal gas).
- Pressure and temperature are intensive; volume and energy are extensive.

Remember: One line: closed system Q = ΔU + W; open system SFEE. Assertion–Reason: (A) PMM-I is impossible. (R) It violates conservation of energy. — Both true; R explains A.

30 Most Important One-Liner Facts — Basic Concepts & First Law of Thermodynamics

1. System with only energy transfer?

— Closed system.

2. Both mass and energy transfer?

— Open system.

3. No transfer at all?

— Isolated system.

4. Another name for open system?

— Control volume.

5. Property independent of mass?

— Intensive property.

6. Volume is which property?

— Extensive.

7. Basis of thermometry?

— Zeroth law.

8. Law of energy conservation?

— First law.

9. First law for closed system?

— Q = ΔU + W.

10. For a cyclic process?

— $\oint \delta Q = \oint \delta W$.

11. Heat and work are?

— Path functions.

12. Internal energy is?

— A property (point function).

13. Enthalpy?

— H = U + pV.

14. $c_p - c_v$ =

— R.

15. γ of air?

— 1.4.

16. R for air?

— 0.287 kJ/kg·K.

17. Work in constant volume process?

— Zero.

18. Heat in adiabatic process?

— Zero.

19. ΔU in isothermal (ideal gas)?

— Zero.

20. Constant in throttling?

— Enthalpy.

21. Work in free expansion?

— Zero.

22. PMM-I violates?

— First law.

23. SFEE stands for?

— Steady Flow Energy Equation.

24. Turbine work (SFEE)?

— $h_1 - h_2$.

25. Nozzle exit velocity?

— $\sqrt{(2\Delta h)}$.

26. SFEE for boiler?

— $q = h_2 - h_1$.

27. 1 cal = ?

— 4.186 J.

28. Isothermal work?

— $p_1V_1 \ln(V_2/V_1)$.

29. Adiabatic work?

— $(p_1V_1 - p_2V_2)/(\gamma - 1)$.

30. Process representable on p–V?

— Quasi-static.

Practice MCQs — Basic Concepts & First Law of Thermodynamics

Q1. A system with energy but no mass transfer is:

(a) Open
(b) Closed
(c) Isolated
(d) Semi-open

✔ **Answer: (b) Closed**

Explanation: Closed system.

Q2. Which is an intensive property?

(a) Volume
(b) Energy
(c) Pressure
(d) Enthalpy

✔ **Answer: (c) Pressure**

Explanation: Independent of mass.

Q3. Heat and work are:

(a) Properties
(b) Point functions
(c) Path functions
(d) Intensive properties

✔ **Answer: (c) Path functions**

Explanation: Depend on the path.

Q4. First law for a closed system:

(a) $Q = \Delta U + W$
(b) $Q = \Delta H + W$
(c) $Q = W$
(d) $Q = \Delta U - W$

✔ **Answer: (a) $Q = \Delta U + W$**

Explanation: Energy conservation.

Q5. Work in a constant volume process:

(a) Maximum
(b) Zero
(c) p ΔV
(d) Infinite

✔ **Answer: (b) Zero**

Explanation: dV = 0.

Q6. In a throttling process, constant is:

(a) Temperature
(b) Pressure
(c) Enthalpy
(d) Entropy

✔ **Answer: (c) Enthalpy**

Explanation: $h_1 = h_2$.

Q7. Thermometry is based on:

(a) Zeroth law
(b) First law
(c) Second law
(d) Third law

✔ **Answer: (a) Zeroth law**

Explanation: Thermal equilibrium.

Q8. Nozzle enthalpy drop 125 kJ/kg — exit velocity (inlet negligible):

(a) 250 m/s
(b) 354 m/s
(c) 500 m/s
(d) 125 m/s

✔ **Answer: (c) 500 m/s**

Explanation: √(2 × 125000) = 500.

Q9. In free expansion:

(a) W = 0, Q = 0
(b) W = 0, Q ≠ 0
(c) W ≠ 0, Q = 0
(d) ΔU ≠ 0

✔ **Answer: (a) W = 0, Q = 0**

Explanation: Expansion into vacuum, ΔU = 0.

Q10. Value of γ for air:

(a) 1.0
(b) 1.33
(c) 1.4
(d) 1.67

✔ **Answer: (c) 1.4**

Explanation: c_p/c_v = 1.005/0.718.

Q11. PMM-I violates:

(a) Zeroth law
(b) First law
(c) Second law
(d) Third law

✔ **Answer: (b) First law**

Explanation: Energy conservation.

Q12. In an isothermal process of an ideal gas:

(a) $Q = 0$ (b) $W = 0$

(c) $\Delta U = 0$ (d) $\Delta H \neq 0$

✔ **Answer: (c) $\Delta U = 0$**

Explanation: U depends only on T.

Q13. Turbine work from SFEE (KE, PE, Q negligible):

(a) $h_1 + h_2$ (b) $h_1 - h_2$

(c) $h_2 - h_1$ (d) $u_1 - u_2$

✔ **Answer: (b) $h_1 - h_2$**

Explanation: Enthalpy drop.

Q14. Total energy of an isolated system:

(a) Increases (b) Decreases

(c) Remains constant (d) Becomes zero

✔ **Answer: (c) Remains constant**

Explanation: No exchange.

Q15. Heat supplied at constant pressure equals change in:

(a) Internal energy (b) Enthalpy

(c) Entropy (d) Work

✔ **Answer: (b) Enthalpy**

Explanation: $Q = \Delta H$.

SSC JE / UPSSSC JE Previous Year Questions — Basic Concepts & First Law of Thermodynamics

Q1. The steady flow energy equation is based on:

SSC JE (Previous Year)

(a) Mass conservation (b) Energy conservation

(c) Momentum conservation (d) Second law

✔ **Answer: (b) Energy conservation**

Explanation: First law.

Q2. When a system completes a cycle:

SSC JE (Previous Year)

(a) $\oint \delta Q = \oint \delta W$ (b) $\oint \delta Q = 0$

(c) $\oint \delta W = 0$ (d) $\oint dU \neq 0$

✔ **Answer: (a) $\oint \delta Q = \oint \delta W$**

Explanation: First law for a cycle.

Q3. Which is not a property? SSC JE (Previous Year)

(a) Pressure
(b) Temperature
(c) Heat
(d) Enthalpy

✔ **Answer: (c) Heat**

Explanation: Heat is a path function.

Q4. In an adiabatic process: SSC JE (Previous Year)

(a) No heat transfer
(b) Constant temperature
(c) Constant pressure
(d) Constant volume

✔ **Answer: (a) No heat transfer**

Explanation: Q = 0.

Q5. Throttling is a ____ process: SSC JE (Previous Year)

(a) Reversible
(b) Irreversible isenthalpic
(c) Isothermal
(d) Isentropic

✔ **Answer: (b) Irreversible isenthalpic**

Explanation: h constant, irreversible.

Q6. Example of an open system: SSC JE (Previous Year)

(a) Gas in a closed cylinder
(b) Steam turbine
(c) Thermos flask
(d) Sealed tank

✔ **Answer: (b) Steam turbine**

Explanation: Mass flows through.

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Chapter 3: Second Law of Thermodynamics

Second Law of Thermodynamics – Basic Concept

The first law only accounts for energy; it does not give direction. The **second law** states that heat flows by itself only from hot to cold and that heat cannot be fully converted into work. The limits of heat engines, heat pumps and refrigerators come from this law.

1. Key Points

- **Thermal reservoir:** a body of such large heat capacity that its temperature does not change on heat exchange. The one supplying heat is the **source**, the one receiving it is the **sink**.
- **Kelvin–Planck statement:** no cyclic engine can receive heat from a single reservoir and convert it completely into work — a 100% efficient heat engine (PMM-II) is impossible.
- **Clausius statement:** heat cannot flow by itself from a colder to a hotter body without external work — a refrigerator needs work input.
- **The two statements are equivalent:** violating one implies violating the other.
- **Heat engine (HE):** takes Q_1 from the source, does work W and rejects Q_2 to the sink; $\eta = W/Q_1 = 1 - Q_2/Q_1$.
- **Refrigerator:** removes Q_2 from the cold space and rejects Q_1 to the hot; $COP_R = Q_2/W = T_2/(T_1 - T_2)$ (Carnot).
- **Heat pump:** aims to supply Q_1 to the hot space; $COP_{HP} = Q_1/W = T_1/(T_1 - T_2)$; and $COP_{HP} = COP_R + 1$.
- **Carnot cycle:** two isothermal and two isentropic (reversible adiabatic) processes; $\eta_{Carnot} = 1 - T_2/T_1$ (T in kelvin). It is the maximum possible efficiency between two given reservoirs (Carnot's theorem).
- **Entropy:** $dS = \delta Q_{rev}/T$; the entropy of an isolated system never decreases ($\Delta S \geq 0$). A reversible adiabatic process is isentropic.
- **Causes of irreversibility:** friction, heat transfer across a finite temperature difference, free expansion, mixing and inelastic deformation.

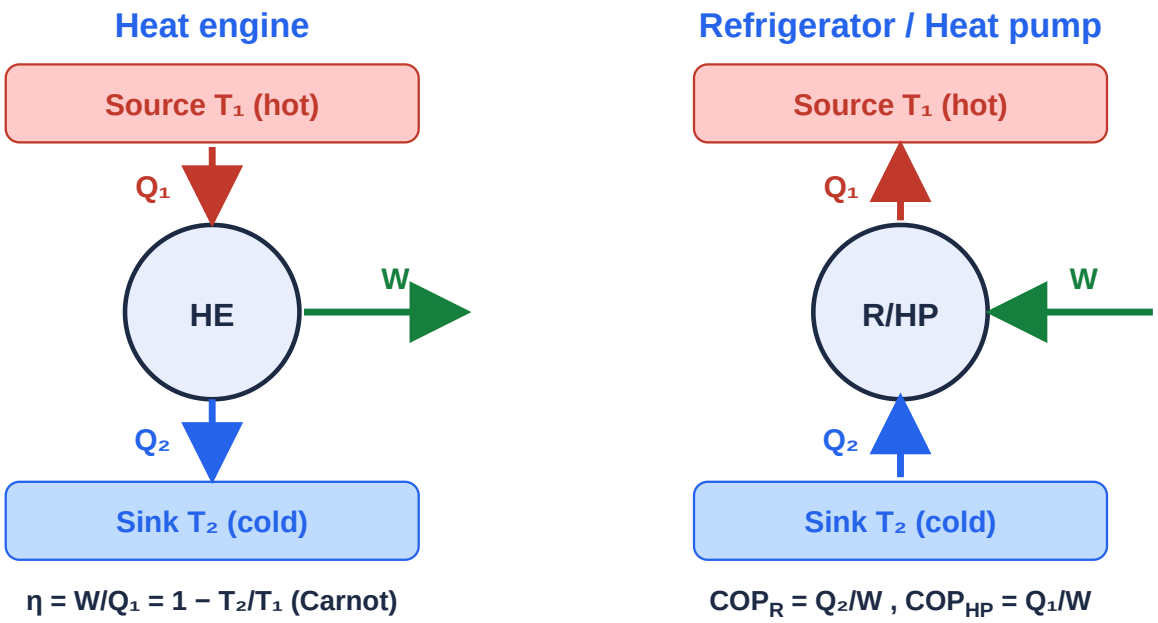


Fig 4.3: Heat engine and refrigerator/heat pump

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Heat engine efficiency	$\eta = W/Q_1 = 1 - Q_2/Q_1$
Carnot efficiency	$\eta = 1 - T_2/T_1$
Refrigerator COP	$COP_R = Q_2/W = T_2/(T_1 - T_2)$
Heat pump COP	$COP_{HP} = Q_1/W = T_1/(T_1 - T_2)$
Relation	$COP_{HP} = COP_R + 1$
Energy balance	$Q_1 = W + Q_2$
Entropy	$dS = \delta Q_{rev}/T$
Clausius inequality	$\oint \delta Q/T \leq 0$
1 ton of refrigeration	≈ 3.5 kW (210 kJ/min)

3. Detailed Explanation

- **Four processes of the Carnot cycle:** (1–2) isothermal expansion at T_1 (Q_1 added), (2–3) isentropic expansion, (3–4) isothermal compression at T_2 (Q_2 rejected), (4–1) isentropic compression. On a T–s diagram it is a rectangle.
- **Corollaries of Carnot's theorem:** (i) no engine between the same reservoirs is more efficient than a reversible one; (ii) all reversible engines have the same efficiency — independent of the working substance.
- **Practical limitation:** the Carnot cycle is not practical, as isothermal processes must be infinitely slow and adiabatic ones infinitely fast; but it is the standard for comparison.

- **Raising efficiency:** increase T_1 or decrease T_2; lowering T_2 is more effective, but the sink is usually the atmosphere, which limits it.
- **Use of a heat pump:** heating rooms in winter; its COP is always greater than 1, so it is cheaper than an electric heater.

4. Comparison Table — Heat Engine, Refrigerator and Heat Pump

Basis	Heat engine	Refrigerator	Heat pump
Purpose	Produce work	Remove heat from cold space	Supply heat to hot space
Work	Output	Input	Input
Performance measure	$\eta = W/Q_1$	$COP = Q_2/W$	$COP = Q_1/W$
Value	Always < 1	> 1 or < 1	Always > 1

5. Solved Numerical Problems

Example 1. A Carnot engine works between 527°C and 27°C. Find its efficiency.

Solution:

1. $T_1 = 800$ K, $T_2 = 300$ K
2. $\eta = 1 - 300/800 =$ **62.5%**

Example 2. An engine receives 1000 kJ and produces 400 kJ of work. Find heat rejected and efficiency.

Solution:

1. $Q_2 = 1000 - 400 =$ **600 kJ**
2. $\eta = 400/1000 =$ **40%**

Example 3. A Carnot refrigerator works between −13°C and 27°C. Find COP_R and COP_{HP}.

Solution:

1. $T_2 = 260$ K, $T_1 = 300$ K
2. $COP_R = 260/40 =$ **6.5**
3. $COP_{HP} = 6.5 + 1 =$ **7.5**

Example 4. A refrigerator removes 5 kW of heat with COP = 4. Find power input and heat rejected.

Solution:

1. $W = Q_2/COP = 5/4 =$ **1.25 kW**
2. $Q_1 = 5 + 1.25 =$ **6.25 kW**

Example 5. An inventor claims an engine working between 400 K and 300 K with 30% efficiency. Is it possible?

Solution:

1. $\eta_{Carnot} = 1 - 300/400 = 25\%$
2. Claim (30%) > Carnot (25%), so **impossible**

6. Frequently Asked Facts

- Carnot efficiency $1 - T_2/T_1$, temperatures in kelvin.
- $COP_{HP} = COP_R + 1$.
- Kelvin–Planck — heat engine; Clausius — refrigerator.
- PMM-II violates the second law.

7. Note (Confusing Facts)

- Use kelvin, not °C, in the Carnot formula — the most common mistake.
- COP can exceed 1; efficiency cannot.
- A reversible engine's efficiency does not depend on the working substance.

Remember: One line: $\eta_{max} = 1 - T_2/T_1$; $COP_R = T_2/(T_1 - T_2)$. Assertion–Reason: (A) A 100% efficient heat engine is impossible. (R) Some heat must be rejected to a sink (Kelvin–Planck). — Both true; R explains A.

30 Most Important One-Liner Facts — Second Law of Thermodynamics

1. Reservoir supplying heat?

— Source.

2. Reservoir receiving heat?

— Sink.

3. Statement about heat engine?

— Kelvin–Planck.

4. Statement about refrigerator?

— Clausius.

5. PMM-II violates?

— Second law.

6. Heat engine efficiency?

— W/Q_1.

7. Carnot efficiency?

— $1 - T_2/T_1$.

8. Processes of Carnot cycle?

— 2 isothermal + 2 isentropic.

9. Shape of Carnot cycle on T–s?

— Rectangle.

10. Refrigerator COP?

— Q_2/W.

11. Heat pump COP?

— Q_1/W.

12. $COP_{HP} - COP_R$?

— 1.

13. Carnot refrigerator COP?

— $T_2/(T_1 - T_2)$.

14. Heat pump COP is always?

— Greater than 1.

15. Engine efficiency is always?

— Less than 1.

16. Max efficiency between two reservoirs?

— Carnot (reversible).

17. Reversible engine efficiency depends on?

— Only reservoir temperatures.

18. Definition of entropy?

— $dS = \delta Q_{rev}/T$.

19. Entropy of isolated system?

— Never decreases.

20. Reversible adiabatic process?

— Isentropic.

21. Unit of entropy?

— kJ/K.

22. Clausius inequality?

— $\oint \delta Q/T \leq 0$.

23. A cause of irreversibility?

— Friction.

24. 1 TR approximately?

— 3.5 kW.

25. Energy balance (HE)?

— $Q_1 = W + Q_2$.

26. More effective way to raise η?

— Lower T_2.

27. Device to heat room in winter?

— Heat pump.

28. Heat flows by itself?

— Hot to cold.

29. Why Carnot impractical?

— Needs infinitely slow/fast processes.

30. Feature of thermal reservoir?

— Temperature stays constant.

Practice MCQs — Second Law of Thermodynamics

Q1. Efficiency of a Carnot engine:

(a) $1 - T_1/T_2$
(b) $1 - T_2/T_1$
(c) T_2/T_1
(d) $T_1/(T_1 - T_2)$

✔ **Answer: (b) $1 - T_2/T_1$**

Explanation: T in kelvin.

Q2. Kelvin–Planck statement relates to:

(a) Refrigerator
(b) Heat engine
(c) Heat pump
(d) Compressor

✔ **Answer: (b) Heat engine**

Explanation: 100% efficient engine impossible.

Q3. Relation of COP_{HP} and COP_R:

(a) $COP_{HP} = COP_R - 1$
(b) $COP_{HP} = COP_R + 1$
(c) $COP_{HP} = 2COP_R$
(d) $COP_{HP} = 1/COP_R$

✔ **Answer: (b) $COP_{HP} = COP_R + 1$**

Explanation: $Q_1 = Q_2 + W$.

Q4. Carnot efficiency between 327°C and 27°C:

(a) 50%
(b) 91.7%
(c) 8.3%
(d) 25%

✔ **Answer: (a) 50%**

Explanation: $1 - 300/600 = 0.5$.

Q5. PMM-II violates:

(a) First law
(b) Second law
(c) Zeroth law
(d) None

✔ **Answer: (b) Second law**

Explanation: Complete conversion from one reservoir.

Q6. The Carnot cycle consists of:

(a) 2 isothermal + 2 isobaric
(b) 2 isothermal + 2 isentropic
(c) 2 isochoric + 2 isentropic
(d) 2 isobaric + 2 isentropic

✔ **Answer: (b) 2 isothermal + 2 isentropic**

Explanation: Carnot.

Q7. COP of a Carnot refrigerator between −3°C and 27°C:

(a) 9
(b) 10
(c) 0.9
(d) 11

✔ **Answer: (a) 9**

Explanation: 270/30 = 9.

Q8. COP of a heat pump is:

(a) Always less than 1
(b) Always more than 1
(c) Equal to 1
(d) Zero

✔ **Answer: (b) Always more than 1**

Explanation: $COP_R + 1$.

Q9. Heat cannot flow from cold to hot without work — this is:

(a) Kelvin–Planck
(b) Clausius
(c) Carnot
(d) Joule

✔ **Answer: (b) Clausius**

Explanation: Clausius statement.

Q10. Entropy of an isolated system in an irreversible process:

(a) Decreases
(b) Increases
(c) Constant
(d) Zero

✔ **Answer: (b) Increases**

Explanation: $\Delta S > 0$.

Q11. Efficiency of an engine taking 1000 kJ and giving 300 kJ work:

(a) 30%
(b) 70%
(c) 43%
(d) 3.3%

✔ **Answer: (a) 30%**

Explanation: 300/1000.

Q12. Efficiency of a reversible engine depends on:

(a) Working substance
(b) Reservoir temperatures
(c) Engine size
(d) Speed

✔ **Answer: (b) Reservoir temperatures**

Explanation: Carnot's theorem.

Q13. 1 ton of refrigeration is approximately:

(a) 1 kW
(b) 3.5 kW
(c) 10 kW
(d) 746 W

✔ **Answer: (b) 3.5 kW**

Explanation: 210 kJ/min ≈ 3.5 kW.

Q14. On a T–s diagram, the Carnot cycle appears as:

(a) Circle
(b) Rectangle
(c) Triangle
(d) Parabola

✔ **Answer: (b) Rectangle**

Explanation: Two horizontal and two vertical lines.

Q15. More effective way to increase heat engine efficiency:

(a) Increase T_1
(b) Decrease T_2
(c) Both equal
(d) Neither

✔ **Answer: (b) Decrease T_2**

Explanation: Efficiency is more sensitive to T_2.

SSC JE / UPSSSC JE Previous Year Questions — Second Law of Thermodynamics

Q1. COP of a heat pump compared to that of a refrigerator is:

SSC JE (Previous Year)

(a) Greater by 1
(b) Less by 1
(c) Equal
(d) Double

✔ **Answer: (a) Greater by 1**

Explanation: $COP_{HP} = COP_R + 1$.

Q2. Efficiency of the Carnot cycle depends on:

SSC JE (Previous Year)

(a) Working substance
(b) Only source and sink temperatures
(c) Pressure
(d) Volume

✔ **Answer: (b) Only source and sink temperatures**

Explanation: $\eta = 1 - T_2/T_1$.

Q3. By the second law, the efficiency of a heat engine is: SSC JE (Previous Year)

(a) Can be 100%
(b) Always less than 100%
(c) More than 100%
(d) Zero

✔ **Answer: (b) Always less than 100%**

Explanation: Kelvin–Planck.

Q4. Entropy is associated with: SSC JE (Previous Year)

(a) Zeroth law
(b) First law
(c) Second law
(d) Newton's law

✔ **Answer: (c) Second law**

Explanation: Second law.

Q5. A refrigerator works on: SSC JE (Previous Year)

(a) Carnot cycle
(b) Reversed Carnot cycle
(c) Otto cycle
(d) Diesel cycle

✔ **Answer: (b) Reversed Carnot cycle**

Explanation: Ideally reversed Carnot.

Q6. Maximum efficiency of an engine between 27°C and 327°C: SSC JE (Previous Year)

(a) 50%
(b) 92%
(c) 8%
(d) 100%

✔ **Answer: (a) 50%**

Explanation: 1 − 300/600.

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